# TrueRehab-RF: what to do first

A = promised in the proposal, not done | B = problems in current hardware and software | as of 2026-11-09

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|---|---|---|---|---|
| **1** First full recording session<br>Radar + MyoWare + IMU/webcam on a person | **2** Run the Pi<br>Flash SD, install, live radar | **3** Decide phase output<br>Phase or micro-vibration from radar? | **4** Fix or drop the IMU<br>Don't let it block step 1 | **5** Ethics before recruiting<br>Consent for healthy volunteers |

## A. Promised in the proposal, not done

ranked by demo and claim dependence

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|---|---|---|
| 1 | **Phase 1: elbow-flexion proof of concept**<br>Radar sees curls; no ground-truth comparison yet | partly |
| 2 | **Phase 2: muscle signal investigation**<br>Radar vs sEMG to failure; MyoWare not yet worn | not done |
| 3 | **Phase 3: compensation detection**<br>Shoulder elevation and trunk lean | not done |
| 4 | **Phase 7: demonstration system on the Pi**<br>Dashboard and bridge built; not run on Pi or live radar | partly |
| 5 | **Phase 4: fusion into 3 scores**<br>Network exists; untrained on real data | code only |
| 6 | **Phase 5: predict compensation in next reps**<br>Needs real rep sequences first | not done |
| 7 | **Phase 6: personal fatigue signature**<br>Needs 3+ people recorded | not done |
| 8 | **Action recommendation (5 actions)**<br>Thresholds invented; no physio review | placeholder |
| 9 | **Micro-Doppler / spectrogram features**<br>Proposal assumed raw I/Q; Plan B has none | blocked |
| 10 | **Force sensor / load cell ground truth**<br>Listed as possible; none acquired | optional |

## B. Problems in current work

ranked by how much they block

| | | |
|---|---|---|
| 1 | **Sensors never recorded together on a body**<br>Blocks A1–A3 and A5–A7 | blocker |
| 2 | **No phase or raw radar output**<br>Ethernet unverified; firmware change needs approval | blocker |
| 3 | **Raspberry Pi never run**<br>SD not flashed; package install needs your OK | blocker |
| 4 | **Model trained on fake data only**<br>Point layers hurt on fake data; 128 frames crops reps | |
| 5 | **IMU wiring fails on the arm**<br>Lost 36 of 72 s; fix or drop | |
| 6 | **Sensor time-sync unproven on real data**<br>Only bench and fake-data tests | |
| 7 | **Radar gaps and unverified left/right**<br>Sequence gaps on long Windows runs | |
| 8 | **sEMG thresholds and labels are guesses**<br>Need a real recording and a physiotherapist | |
| 9 | **Ethics and consent undocumented**<br>Healthy-volunteer approval; model needs 3+ people | |
| 10 | **Deck, brochure, housekeeping**<br>Deck out of date; brochure not started; unpushed files | |

**Honesty note:** proposal fatigue and risk percentages are illustrative. Until step 1 is done, present this as a working acquisition and dashboard platform, not validated fatigue or compensation detection.